

**Post-MDT Oncology Workflow
Bottlenecks: A Design-Science Study
of an NHS Integrated Data Repository
and Decision-Support Dashboard to
Reduce Delays from MDT Outcome to
Treatment Start.**

Mark Collins

A dissertation submitted in partial fulfilment
of the requirements for the degree of
Master of Science (MSc) in Enterprise IT Management

July 28, 2026

Contents

1	Introduction	4
1.1	Problem Statement	4
1.2	Research Aim and Objectives	5
1.3	Research Approach	6
1.4	Academic and Practical Significance	6
2	Literature Review	8
2.1	Introduction	8
2.2	The NHS Cancer Pathway and MDT Framework	10
2.3	Operational Delays in Cancer Pathways	11
2.4	Data Fragmentation and the Need for Integrated Data Repositories	13
2.5	Dashboards, Operational Analytics, and Decision-Support Systems	15
2.6	Usability, HCI, and ISO 9241-110	17
2.7	AI-Enabled MDT Systems and Emerging Technologies	17
2.8	Synthesis of Gaps	18
2.9	Summary	19
3	Methodology	21
3.1	Overview of Research Design	21
3.2	Design Science Research Framework	22
3.3	Design and Development	23
3.4	Design Principles	30
3.5	Evaluation Strategy	32
3.6	Ethical and Governance Considerations	32
3.7	Summary	33
4	Results and Analysis	34
4.1	Introduction	34
4.2	Repository Development Results	34
4.3	Oncology Referral Analysis	35
4.4	Radiotherapy Service Analysis	36
4.5	End-to-End Pathway Analysis	38

4.6	Evaluation of Design Principles	40
4.7	Summary	40
References		41
List of Acronyms		46

DRAFT

Chapter 1

Introduction

Delivering timely cancer care remains a critical priority within the NHS, where delays at any stage of the diagnostic and treatment pathway can negatively affect patient outcomes, experience, and service performance. Recent policy reforms have consolidated the NHS Cancer Waiting Time (CWT) framework into three outcome-focused standards: the 28-day Faster Diagnosis Standard (FDS), the 31-day decision-to-treat to treatment (DTT) standard, and the 62-day referral-to-treatment (RTT) standard—implemented nationally from October 2023 (Johnson, 2024; NHS Digital, 2026). These standards place increased emphasis on operational viability, rapid decision-making, and efficient progression of patients through the post-MDT (multidisciplinary team) stages of the oncology pathway. However, many Trusts continue to face systemic challenges in generating timely, accurate, and actionable insights from fragmented data sources.

Within Portsmouth Hospitals University NHS Trust, as in many NHS institutions, key operational data concerning MDT outcomes, clinic scheduling, booking practices, and treatment initiation remain dispersed across siloed systems. Although these systems individually support MDT coordination, outpatient clinic management, and treatment delivery, they lack a consolidated view of pathway performance. This fragmentation makes it difficult for operational managers to detect bottlenecks, identify emerging breach risks, or allocate clinical capacity efficiently. As a result, delays between **MDT outcomes**, **first oncology clinic**, **decision-to-treat**, and **first definitive treatment** remain insufficiently visible, impairing the Trust’s ability to monitor compliance with the 31- and 62-day standards and to respond rapidly to operational pressures.

1.1 Problem Statement

The central problem addressed in this research is the absence of an integrated, operationally focused data infrastructure and decision-support interface to illuminate post-MDT workflow delays and guide timely interventions within an NHS oncology service. Despite the availability of relevant data, the lack of integration and harmonisation across MDT, clinic, and treatment

systems results in manual workarounds, inconsistent performance monitoring, and delayed insights. This gap diminishes the ability of operational teams to identify patients at risk of breaching national cancer standards, understand the factors driving variability in pathway intervals, and proactively manage capacity-constrained environments.

Moreover, existing reporting solutions tend to focus on aggregate compliance metrics after the fact, rather than providing real-time or near-real-time visibility of operational bottlenecks. Without a unified source of truth that aligns event definitions, timestamps, and clinical coding standards (NHS Digital, 2026), it becomes difficult to execute performance analysis or support day-to-day decision-making. Literature on Integrated Data Repositories (IDRs) similarly highlights that many institutions struggle with data harmonisation, terminological alignment, and the creation of analytics-ready architectures capable of supporting both research and operational use cases (Gagalova *et al.*, 2020).

This study therefore seeks to design, develop, and evaluate a **lightweight Integrated Data Repository (IDR)** and accompanying **decision-support dashboard** tailored specifically to the post-MDT oncology workflow. By integrating key pathway events, clinic capacity information, and national standard definitions, the artefact aims to surface bottlenecks, highlight breach risks, and support operational teams in prioritising actions that improve pathway performance.

1.2 Research Aim and Objectives

1.2.1 Research Aim

To design and evaluate an Integrated Data Repository and decision-support dashboard capable of improving operational visibility and proactive management of post-MDT oncology pathways.

1.2.2 Objectives

1. Identify and quantify post-MDT pathway delays.
2. Determine operational factors associated with delay.
3. Design an integrated data architecture.
4. Develop a user-centred dashboard.
5. Evaluate utility, usability and technical validity.
6. Derive transferable design principles

Research Question	DSR Activity	Primary Methods
RQ1: Where do delays occur between MDT outcome and treatment?	Problem identification	Descriptive statistics, interval analysis
RQ2: What operational factors predict extended delays?	Problem analysis	Regression and capacity analysis
RQ3: Does the artefact improve operational decision-making?	Evaluation	Task-based testing, time-to-answer, accuracy
RQ4: What design principles support post-MDT operational analytics?	Design knowledge	Literature synthesis and evaluation synthesis

Table 1.1: This mapping ensures methodological coherence between the research questions, artefact, and evaluation.

1.3 Research Approach

This study adopts Design Science Research (DSR) because the primary objective is not solely to explain an existing phenomenon but to create and evaluate an artefact intended to address an identified organisational problem. Whereas case-study research focuses on understanding contextual factors and Action Research emphasises organisational change through iterative intervention, DSR provides a structured framework for the design, construction and evaluation of innovative technological artefacts while simultaneously generating transferable knowledge (Hevner *et al.*, 2004; Peffers *et al.*, 2007).

The approach is particularly appropriate given the need to integrate heterogeneous operational data sources and translate them into actionable decision support for NHS oncology services. Similar DSR approaches have been successfully applied within health informatics, clinical decision support, and healthcare analytics domains where practical solutions and theoretical contribution are required concurrently.

The study therefore follows the six-stage Design Science Research Methodology (DSRM) proposed by Peffers *et al.* (2007), with detailed application described in Chapter 3.

1.4 Academic and Practical Significance

Academically, the research contributes to the limited body of DSR-grounded studies focusing on NHS operational analytics, offering:

- an IDR reference architecture for oncology operations;
- a set of design principles for NHS-appropriate decision-support dashboards;
- an evaluation protocol combining usability metrics, decision-quality testing, and technical verification.

Practically, the artefact provides operational teams with a tool that can:

- reduce time-to-insight;
- support rapid identification of bottlenecks and breach risks;
- align capacity planning with demand;
- improve pathway monitoring in line with the 28-, 31-, and 62-day standards.

Ultimately, the study aims to support faster, more reliable progression from MDT decision to treatment initiation, with the potential to support improved patient care and more effective management of oncology services.

DRAFT

Chapter 2

Literature Review

2.1 Introduction

The purpose of this literature review is to synthesise and critically evaluate existing research relevant to the design of an Integrated Data Repository (IDR) and operational analytics dashboard aimed at improving post-MDT oncology pathway efficiency. Timely cancer treatment has profound implications for survival, patient experience, resource utilisation, and system performance. Evidence from national datasets, observational studies, systematic reviews, and informatics research shows that delays between MDT decisions and treatment initiation remain persistent across the NHS. Furthermore, data fragmentation, inconsistent recording practices, and limited visibility of operational bottlenecks hinder the ability of clinicians, MDT coordinators, and service managers to proactively manage these delays.

The review is organised around five interrelated themes:

1. MDT operations and evolving pressures on multidisciplinary cancer decision-making;
2. Operational delays across the cancer pathway, particularly diagnosis-to-treatment intervals (DTI);
3. Data fragmentation and the case for integrated data repositories;
4. Dashboards and performance-monitoring tools to support operational decision-making;
5. Emerging data trends and the role of ETL, interoperability, AI and multimodal data integration.

Collectively, this literature forms the conceptual foundation (kernel theory) for the artefact developed in this study. The following sections provide a detailed account of these themes, integrating high-quality evidence from the UK, international datasets, clinical oncology, health informatics, and organisational science.

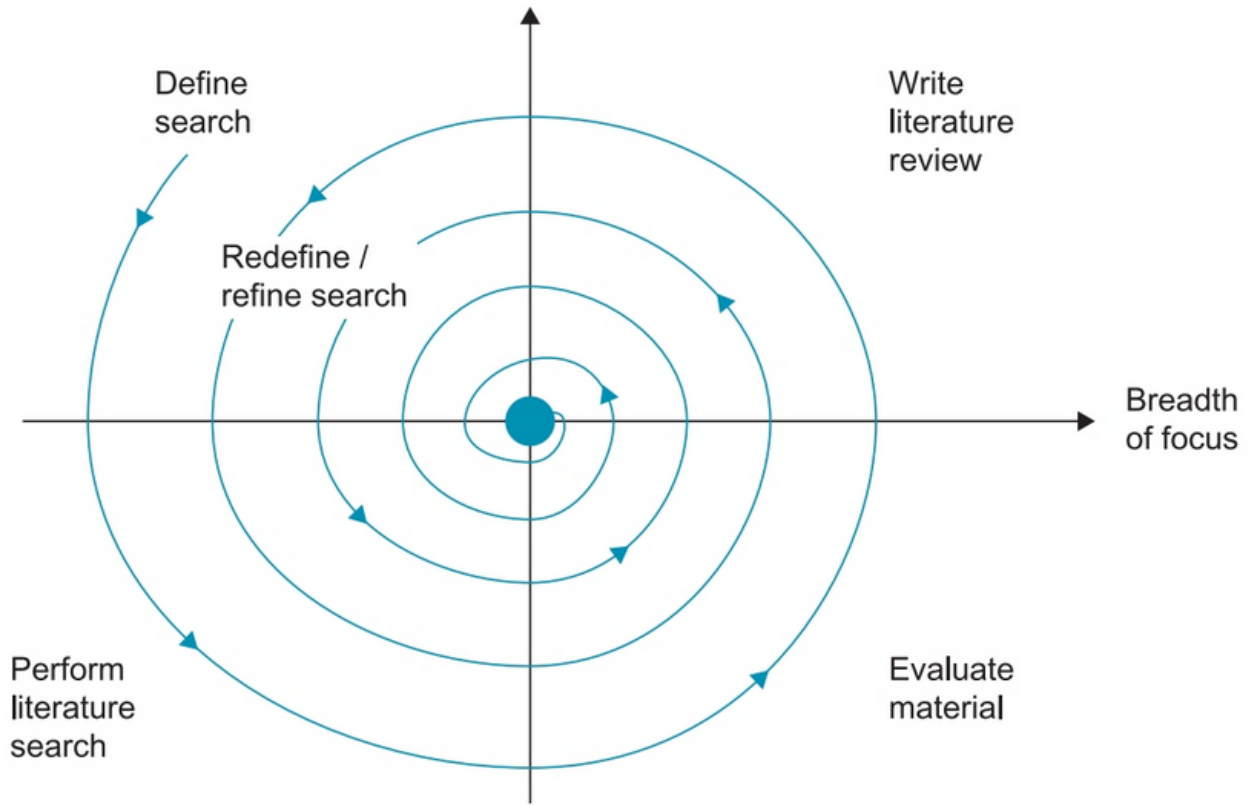


Figure 2.1: Iterative literature review process (Dawson, 2015)

2.1.1 Method

An iterative literature review was adopted, following the guidance outlined by Dawson (2015) (Figure 2.1). An initial search was conducted using Google Scholar and open-access repositories to identify foundational concepts and terminology related to Integrated Data Repositories, cancer pathways, MDT operations, dashboards, and Design Science Research. These preliminary results informed a more structured search strategy applied to ScienceDirect, IEEEXplore, and PubMed.

Searches were limited to peer-reviewed journals, systematic reviews, national audits, and authoritative policy documents published primarily between 2010 and 2026, with earlier seminal works retained where conceptually relevant (e.g. JCCO report (Ash *et al.*, 2004), the Calman-Hine report (Calman and Hine, 1995) and Design Science Research foundations). Studies were included if they addressed MDT processes, oncology pathway delays, health data integration, dashboards, usability standards, or operational analytics. Articles focused exclusively in molecular biology, single-modality clinical trials, or non-healthcare data systems were excluded.

Approximately 90 sources were reviewed during the process, with around 55 core studies retained for detailed analysis based on relevance, methodological quality, and applicability to NHS operational contexts. Backward citation tracking was used to identify influential seed papers and ensure conceptual completeness.

2.2 The NHS Cancer Pathway and MDT Framework

2.2.1 MDTs as the Backbone of UK Cancer Care

Since the publication of the Calman–Hine report (Calman and Hine, 1995) and subsequent interventions, MDTs have been embedded as the central governance structure for cancer decision-making (Haward, 2006; Price *et al.*, 2014; Morris *et al.*, 2006, 2008). The MDT model mandates that every new cancer case in the UK is discussed by a multidisciplinary panel including oncologists, surgeons, radiologists, pathologists, specialist nurses, and allied professionals. The 2024 NHS Wales MDT Charter emphasises the crucial role MDTs play in standardising care, reducing unwarranted variation, and improving outcomes by bringing domain experts together in real time (NHS Wales, 2024). Prospective benefits include higher adherence to guidelines, improved communication, shared decision-making, and better coordination of multi-modal treatment plans.

However, the complexity of modern cancer care has increased dramatically alongside rising incidence, new treatment options, biomarker-based stratification, digital imaging volume, and genomic data availability. MDTs now contend with substantially larger case numbers, more complex diagnostic information, and resource pressures that stretch meeting capacity. Professional societies and national audits consistently highlight that MDT meetings are often over-subscribed, under-resourced, and administratively burdensome. These pressures compromise the quality of discussion and create operational bottlenecks that directly affect patient pathways.

2.2.2 Evidence of MDT Operational Pressures

Collectively, the literature demonstrates that MDTs are experiencing growing operational pressure resulting from increasing cancer incidence, expanding diagnostic complexity, and constrained service capacity. Across multiple studies, common themes emerge including administrative inefficiencies, information deficits, scheduling pressures, and resource limitations.

Soukup *et al.* (2022) identified logistical challenges affecting almost one-third of MDT discussions, while Law *et al.* (2024) highlighted persistent barriers relating to communication, attendance, information availability, and workload. These findings suggest that operational inefficiencies are not isolated incidents but recurring characteristics of contemporary MDT environments.

Evidence from pathway-performance studies reinforces this position. Lim *et al.* (2025) reported poor national compliance with cancer waiting-time standards across UK head and neck services, while Chakravarty *et al.* (2023) demonstrated that additional complexity within referral pathways was associated with significantly longer treatment delays. Together, these studies indicate that operational challenges extend beyond MDT meetings themselves and influence downstream stages of the cancer pathway.

A recurring theme across the literature is therefore the lack of visibility between MDT

decision-making and subsequent treatment delivery. While studies identify operational pressures and poor pathway performance, relatively little attention has been given to how integrated operational data might support earlier identification of emerging bottlenecks.

While these studies consistently demonstrate operational strain within MDT processes, several limitations should be acknowledged. Much of the existing evidence is observational and focuses primarily on meeting effectiveness, communication quality, or decision-making processes rather than downstream operational consequences of MDT outcomes. For example, Soukup *et al.* (2022) identify logistical and administrative barriers that impair MDT performance, yet the study does not examine whether these issues translate into measurable delays in treatment initiation. Similarly, the scoping review proposed by Law *et al.* (2024) aggregates a broad range of organisational challenges but provides limited insight into which barriers exert the greatest influence on pathway performance.

Large-scale cohort studies such as Lim *et al.* (2025) quantify poor compliance with national standards, but their retrospective design offers little visibility of the specific workflow stages responsible for delay accumulation. Consequently, while the literature establishes that MDT processes operate under significant pressure, a gap remains in understanding how operational data can be used to identify, monitor, and mitigate delays arising after MDT decisions have been made. This distinction is important because improving MDT discussion quality alone may be insufficient if subsequent scheduling and treatment processes remain poorly coordinated.

2.2.3 National policy direction and waiting time standards

NHS England’s 2024/25 planning guidance emphasises compliance with the Faster Diagnosis Standard (FDS) and cancer treatment timeliness metrics. Persistent waiting-time challenges, data-quality requirements, and the need for integrated pathway oversight tools are highlighted (Parker, 2026). Radiotherapy services face particular strain: the Royal College of Radiologists reported over 2,300 breaches of the 31-day standard in October 2023 (RCR, 2024). Hidden waits—such as 20-week delays between surgery and radiotherapy—are not systematically captured, further illustrating limitations in operational visibility.

2.3 Operational Delays in Cancer Pathways

2.3.1 Diagnosis-to-Treatment Interval (DTI) as a Critical Metric

A robust body of literature demonstrates that prolonged DTI adversely affects survival across multiple cancer types.

Key findings include:

- Hanna *et al.* (2020): A systematic review of 34 studies (1.27M patients) found each four-week delay increased mortality by up to 6–8% across treatment modalities.

- Coca-Pelaz *et al.* (2018): Delays in Head and Neck Small Cell Carcinoma beyond four weeks contribute to tumour progression and reduced local control.
- Sharma *et al.* (2016): Delays greater than 30 days reduce overall survival (HR 1.12), with mortality increasing 2.2% per additional week.
- Villemure-Poliquin *et al.* (2025): Treatment within 30 days improves survival by 9%.
- Saleh *et al.* (2024): Cervical cancer survival drops dramatically after 30 days.

Despite differences in cancer type, geography, and study design, the evidence consistently demonstrates that treatment delay is associated with poorer clinical outcomes. Systematic reviews and large cohort studies identify remarkably similar trends, with survival deteriorating as diagnosis-to-treatment intervals increase. However, the magnitude of effect varies between tumour groups, suggesting that pathway urgency may differ according to tumour biology and disease progression rates. This consistency across studies strengthens the argument that reducing operational delay is clinically important, while variation in effect size highlights the need for pathway-specific operational management strategies.

2.3.2 Hospital/System Factors Influencing Delay

Across the reviewed studies, organisational influences on delay can be grouped into three broad categories: diagnostic bottlenecks, administrative coordination challenges, and capacity constraints. Diagnostic delays frequently arise from imaging and biopsy availability, while administrative delays result from scheduling inefficiencies, incomplete information, or pathway handovers. Capacity-related delays emerge where demand exceeds available clinic, theatre, chemotherapy, or radiotherapy resources. Although studies differ in emphasis, all point towards organisational processes rather than purely clinical factors as major contributors to pathway delay.

A further limitation of the current literature is the tendency to treat organisational performance as a static characteristic rather than a dynamic process. Factors such as hospital volume, resource availability, and service configuration are frequently associated with treatment delays; however, there is comparatively little exploration of how these risks evolve over time or how operational teams can intervene before delays become breaches. This reinforces the distinction between retrospective performance measurement and proactive pathway management.

2.3.3 MDT-specific drivers of post-decision delays

- Missing diagnostic information and administrative errors cause re-discussion delays (Soukup *et al.*, 2022).
- High caseloads reduce decision completeness.

- Secondary-care referrals introduce additional diagnostic delays (Chakravarty *et al.*, 2023).
- Capacity constraints (radiotherapy, theatres, oncology clinics) generate downstream bottlenecks.

Although the evidence linking treatment delay and mortality is compelling, caution should be exercised when applying these findings directly to operational improvement initiatives. Most studies evaluate aggregated diagnosis-to-treatment intervals and therefore provide limited insight into the relative contribution of individual pathway stages. While the findings clearly establish delay as clinically important, they do not identify where operational interventions are likely to deliver the greatest benefit. From a service management perspective, this highlights the need for more granular operational analytics capable of decomposing pathway delays into actionable components.

2.3.4 Critical synthesis of delay literature

Collectively, the literature provides strong evidence that prolonged diagnosis-to-treatment intervals adversely affect patient outcomes across multiple cancer types. However, important limitations remain when attempting to translate this knowledge into operational practice. Most studies conceptualise delay as a single aggregated interval and consequently offer limited support for identifying specific workflow bottlenecks. Furthermore, while organisational variables such as hospital volume, diagnostic complexity, and service capacity are frequently identified as contributors to delay, relatively little attention has been given to the role of integrated operational data in supporting proactive intervention.

This omission is particularly significant within NHS oncology services, where national performance standards require action before breaches occur. Once a patient has experienced a clinically significant delay, retrospective performance reporting provides little opportunity to prevent harm. Accordingly, there is a need for analytical approaches capable of identifying emerging risks and supporting early operational intervention, rather than simply documenting historical performance.

2.4 Data Fragmentation and the Need for Integrated Data Repositories

2.4.1 Fragmentation Across NHS Systems

Oncology data spans EPRs, PAS, pathology, RIS/PACS, radiotherapy systems, chemotherapy e-prescribing, Somerset/Cosmic, and MDT spreadsheets.

Evidence includes:

- Operational policy from Royal Wolverhampton highlights inconsistent event definitions and PTL challenges (The Royal Wolverhampton NHS Trust, 2024).

- Varma *et al.* (2026): 92.3% agreement between CWT and hospital data only when treatment occurs within 100 days; discrepancies arise from definition mismatches and missing fields.

2.4.2 Clinical Data Warehouses (CDWs) and the Evolution Toward IDRs

Examples:

- ROOT CDW: real-time repository for 67,000+ cancer patients (Jung *et al.*, 2021).
- ONCO-FAIR: chemotherapy-specific FHIR extensions (Guilbert *et al.*, 2024).
- NHS-industry multimodal genomic-clinical data lake (Conway *et al.*, 2025).
- HN-XNAT: federated imaging and radiotherapy repository (Butterworth *et al.*, 2025).

Despite differing objectives, several common characteristics can be observed across these repository implementations. All seek to integrate heterogeneous oncology datasets, establish consistent data models, and improve data accessibility for downstream analysis. However, important differences also exist. ROOT prioritises real-time operational data aggregation, whereas ONCO-FAIR focuses on interoperability and standards compliance through FHIR extensions. The NHS-industry data lake emphasises large-scale multimodal research integration, while HN-XNAT adopts a federated model centred on imaging and radiotherapy data.

Collectively, these implementations demonstrate that successful repositories require data harmonisation, governance frameworks, and scalable integration architectures. However, few have been designed specifically to support operational cancer pathway management. This distinction directly informs the proposed artefact, which prioritises workflow visibility, breach-risk identification, and operational decision support rather than research-oriented data reuse.

While these examples demonstrate the technical feasibility of large-scale oncology data repositories, their transferability to routine NHS operational environments warrants careful consideration. Many reported implementations originate from digitally mature organisations with substantial informatics capability, dedicated technical teams, and highly structured data environments. Such conditions may not be representative of NHS Trusts where operational data are frequently distributed across legacy systems, locally developed databases, spreadsheets, and departmental applications.

Furthermore, the majority of published repositories prioritise research, population health, or precision medicine objectives. Although these use cases require extensive data integration, they often place less emphasis on the timeliness, operational responsiveness, and workflow visibility required for day-to-day pathway management. Consequently, a gap exists between the design of many clinical data repositories and the practical needs of operational cancer services.

2.4.3 Need for IDRs in NHS Operational Oncology

The absence of a robust IDR has implications beyond simple reporting inefficiency. Fragmented datasets create multiple versions of the truth, requiring operational staff to reconcile conflicting information from different systems before decisions can be made. This process introduces delays, increases administrative burden, and reduces confidence in performance measures.

Moreover, fragmented data environments encourage reactive management practices. Operational teams frequently identify pathway risks only after breaches have occurred because no single system provides a comprehensive view of patient progression across MDT, clinic, and treatment stages. As a result, opportunities for early intervention may be missed despite the underlying data already existing within organisational systems.

2.5 Dashboards, Operational Analytics, and Decision-Support Systems

2.5.1 The Role of Dashboards in Performance Management

Dashboards synthesise high-volume data for actionable insight. NHS guidance emphasises dashboards as essential for meeting FDS and treatment targets.

Buttigieg *et al.* (2017) identify:

- Strategic dashboards,
- Tactical dashboards,
- Operational dashboards.

Key design principles include real-time refresh, drill-downs, user-centred design, and integration with data infrastructures.

Despite widespread adoption, dashboards are not universally associated with improved organisational performance. Several studies have reported that dashboards frequently become passive reporting tools rather than active decision-support systems. The mere presentation of information does not guarantee action, particularly when users are confronted with excessive metrics, poorly aligned visualisations, or data that lack clear operational relevance.

Consequently, dashboard effectiveness depends not only on technical implementation but also on the extent to which information is embedded within operational decision-making processes. This distinction is particularly relevant in cancer pathway management where users require actionable intelligence capable of informing scheduling, capacity allocation, and escalation decisions.

2.5.2 Clinical Decision-Support Dashboards and Multimodal Data

Examples:

- Multimodal CDSS supporting 170,000+ patients, using ETL + NLP + 140 data quality checks (Chang *et al.*, 2025).
- ROOT CDW: automated ingestion, structured/unstructured data integration (Jung *et al.*, 2021).

2.5.3 Comparative Analysis of Dashboard Implementations

The reviewed dashboard implementations share several common design objectives, including improved information visibility, support for timely decision-making, and reduction of administrative burden. Both ROOT and multimodal CDSS environments rely upon integrated data architectures to provide trustworthy analytics, emphasising the dependency between dashboard performance and underlying data quality.

However, notable differences exist regarding intended usage. Clinical decision-support dashboards are primarily oriented towards patient-level treatment decisions, whereas operational dashboards focus on service performance, capacity management, and workflow optimisation. The latter perspective receives comparatively less attention within the literature despite its importance for achieving national waiting-time standards.

These observations suggest that an effective oncology operational dashboard should combine characteristics of both approaches: robust data integration, user-centred visualisation, and proactive decision-support capabilities tailored to pathway management.

2.5.4 Data Quality and Informatics Challenges

Dashboards depend on strong data foundations:

- Coding variation limits dataset reliability (Varma *et al.*, 2026).
- Oncology workloads require custom FHIR extensions (Guilbert *et al.*, 2024).
- Multimodal data lakes require governance and standardisation (Conway *et al.*, 2025).

2.5.5 Reactive versus Proactive Operational Management

Traditional healthcare performance reporting has largely focused on retrospective measurement through key performance indicators and compliance monitoring. While such approaches provide valuable accountability mechanisms, they are inherently limited in their ability to prevent adverse outcomes because they identify problems only after they have occurred.

This limitation is particularly relevant in oncology services. Evidence demonstrates that delays in treatment initiation are associated with poorer patient outcomes (Hanna *et al.*, 2020;

Sharma *et al.*, 2016). Consequently, a management approach based solely on retrospective review is problematic because a missed cancer target represents more than a performance failure; it may indicate a clinically significant delay experienced by a patient.

Continuous improvement frameworks such as Plan-Do-Check-Act (PDCA) remain widely used within healthcare quality improvement. However, PDCA assumes that outcomes can be observed, measured, and refined through successive iterations. In cancer pathway management, allowing delays to occur before evaluating performance may be ethically and operationally undesirable. Instead, there is a need for leading indicators capable of identifying emerging risks before breaches occur.

From an operational analytics perspective, this shifts the emphasis from historical reporting towards predictive and proactive decision support. Dashboards designed around breach-risk identification, remaining time-to-target, and capacity constraints may therefore provide greater practical value than systems focused exclusively on retrospective compliance measurement.

2.6 Usability, HCI, and ISO 9241-110

ISO 9241-110:2020 (International Organization for Standardization, 2020) provides seven principles:

1. Suitability for the task
2. Self-descriptiveness
3. Conformity with user expectations
4. Learnability
5. Controllability
6. Error tolerance
7. Suitability for individualisation

Fraunhofer usability research supports its relevance in healthcare (Hunkirchen).

2.7 AI-Enabled MDT Systems and Emerging Technologies

Examples include:

- EvoMDT: multi-agent MDT recommender reducing decision time by 30–40% (Liu *et al.*, 2026a).

- TrustedMDT: TNM staging and guideline checking integrated with Teams (Soltan *et al.*, 2025).
- AI-MDT for lung cancer: improved throughput and workload reduction (Liu *et al.*, 2026b).
- AI automation of transcription and summarisation (HealthOrbit, 2025).
- Multi-agent consensus MDT system achieving high inter-agent agreement (Han *et al.*, 2025).

While AI-enabled MDT systems report promising improvements in efficiency, consistency, and decision support, many studies remain limited by controlled implementation environments and highly structured datasets. Questions therefore remain regarding external validity and scalability within routine NHS settings characterised by fragmented information systems and variable data quality.

Importantly, the effectiveness of AI applications appears dependent upon the existence of integrated, standardised, and high-quality data architectures. Consequently, AI may be better viewed as a downstream capability rather than a standalone solution. Without addressing foundational challenges relating to interoperability, governance, and data integration, many of the anticipated benefits of AI-assisted MDT decision-making may remain difficult to realise in practice.

2.8 Synthesis of Gaps

The literature reveals a common pattern across otherwise diverse research domains. Studies examining MDT performance identify increasing operational pressure but provide limited visibility of downstream workflow delays. Research linking treatment delays to survival demonstrates the importance of timely treatment but offers limited guidance regarding how delays should be managed operationally. Similarly, data repository research demonstrates integration capability while dashboard studies emphasise information presentation; however, few studies combine these strands into a coherent operational solution.

The literature review identifies four principle gaps.

Gap 1: Limited visibility of post-MDT workflow delays. While existing studies identify treatment delays and MDT inefficiencies, few provide mechanisms for analysing delay accumulation between MDT outcome and treatment initiation. This gap directly informs Objective 1 (identify and quantify post-MDT pathway delays) and Objective 2 (determine operational factors associated with delay).

Gap 2: Predominantly reactive performance management. Most existing reporting approaches focus on retrospective compliance measurement rather than proactive intervention. This gap informs DP1 (Surface Breach Risks as a Leading Indicator) and DP2 (Pair Risk Information with Actionable Capacity Data).

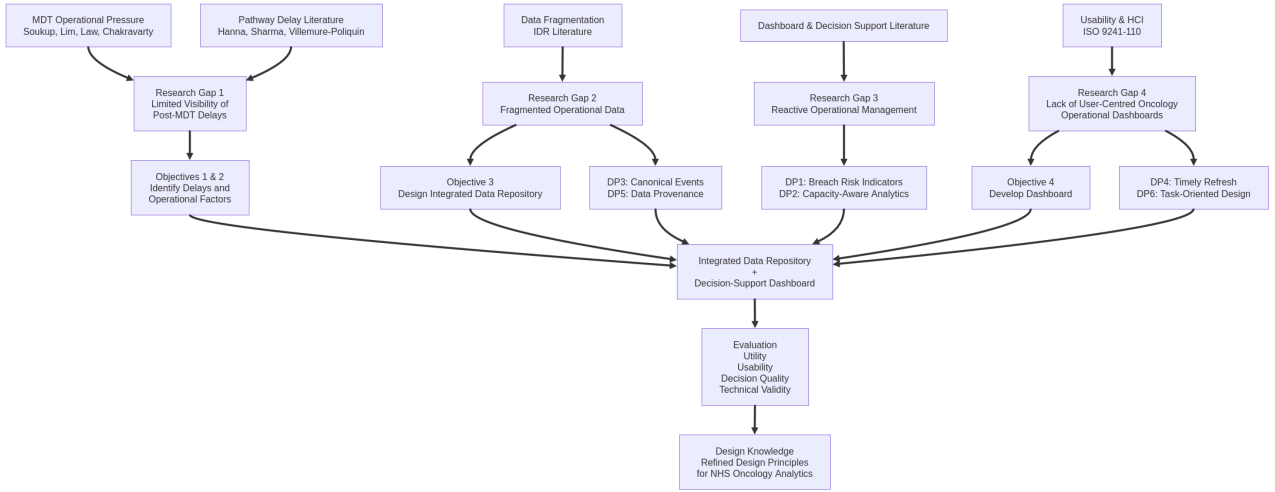


Figure 2.2: Conceptual model linking literature-derived gaps, research objectives, design principles, artefact development, and evaluation

Gap 3: Limited operational application of Integrated Data Repositories. Existing repository research focuses primarily on research, genomics, and population health use cases rather than pathway operations. This gap directly supports Objective 3 (design an integrated data architecture) and DP3 (Use Canonical Standard-Aligned Pathway Events).

Gap 4: Insufficient evidence regarding user-centred operational dashboards. Although dashboard research demonstrates value for healthcare monitoring, limited attention has been given to oncology pathway management and task-oriented operational decision support. This gap informs Objective 4 (develop a user-centred dashboard) and DP6 (Align Interaction Design with Real Operational Tasks).

Collectively, these gaps establish the need for an artefact that combines integrated operational data, proactive pathway monitoring, and user-centred decision support. These requirements form the foundation of the Design Science Research approach outlined in Chapter 3.

Figure 2.2 synthesises the findings of the literature review and illustrates how the identified research gaps inform the objectives and design principles underpinning the artefact. The model demonstrates that challenges relating to MDT operational pressures, pathway delays, data fragmentation, reactive management practices, and usability deficiencies collectively motivate the development of an Integrated Data Repository and decision-support dashboard. These gaps are translated into explicit objectives and design principles which guide artefact development and subsequently form part of the evaluation framework. The resulting design knowledge contributes both a practical solution for NHS oncology services and transferable principles for operational analytics in healthcare.

2.9 Summary

This literature review synthesises evidence across MDT functioning, treatment delays, operational workflow challenges, data fragmentation, health informatics, dashboard design, usability

standards, and emerging AI-driven MDT tools. MDTs remain vital but increasingly pressured. Delays between diagnosis, MDT decisions, and treatment initiation significantly worsen outcomes, yet Trusts lack the integrated systems required to manage these pathways proactively. Modern CDWs demonstrate feasible architectures, while dashboard and usability literature highlight design imperatives. AI-enabled MDT systems reveal future potential but highlight the foundational need for high-quality integrated data repositories.

The combined evidence exposes a clear research gap: NHS oncology services require a robust, user-centred, IDR-backed operational dashboard to improve MDT-to-treatment coordination.

DRAFT

Chapter 3

Methodology

3.1 Overview of Research Design

This research adopts a Design Science Research (DSR) methodology to address an operational problem in NHS oncology services: the lack of integrated, actionable visibility of delays between MDT outcomes and treatment initiation. DSR is particularly suitable for this study because it explicitly focuses on the design, construction, and evaluation of an artefact intended to improve a real-world organisational context, while generating transferable design knowledge (Hevner *et al.*, 2004).

The primary artefact developed in this study is a lightweight-integrated data repository (IDR) and decision-support dashboard, designed to consolidate post-MDT pathway data and support operational decision-making. In addition to the artefact itself, the research produces a set of evaluated design principles for oncology operational analytics, constituting a contribution to knowledge beyond the local implementation.

3.1.1 Justification for Design Science Research

Several methodological approaches were considered. A case study methodology would have enabled detailed examination of post-MDT operational processes but would not have provided a structured framework for the design and evaluation of an intervention. Action Research was also considered because of its focus on organisational change and practitioner involvement. However, Action Research typically requires repeated cycles of intervention and measurement within live operational environments. In cancer pathway management, allowing breaches to occur in order to evaluate an intervention would be ethically and operationally undesirable given the known relationship between treatment delays and patient outcomes.

Design Science Research was therefore selected because it provides a rigorous framework for designing, developing, and evaluating an artefact intended to solve a real-world organisational problem while simultaneously generating transferable knowledge. DSR has been widely applied within healthcare informatics, clinical decision support, and data integration research where new technological artefacts are required to address operational challenges (Hevner *et al.*, 2004;

Peffers *et al.*, 2007).

Unlike traditional continuous-improvement approaches such as Plan-Do-Check-Act (PDCA), cancer pathway management cannot rely solely on post hoc measurement of outcomes. A breach of the 31-day or 62-day standard represents a clinically meaningful delay that may adversely affect patient outcomes. Consequently, operational management must emphasise proactive identification and mitigation of risk rather than retrospective evaluation alone. The design objective of the proposed dashboard is therefore not simply to identify breaches after they occur, but to provide leading indicators that enable intervention before delays become clinically significant.

Case study research was rejected because the objective of this study extends beyond understanding operational delays to designing and evaluating a technological intervention intended to address them. Similarly, Action Research was considered unsuitable because it emphasises organisational change through repeated intervention cycles, whereas the primary contribution of this study is the creation and evaluation of an information systems artefact.

3.2 Design Science Research Framework

The study follows the Design Science Research Methodology (DSRM) proposed by Peffers *et al.* (2007), which structures DSR into six interrelated activities. These activities are applied iteratively rather than strictly linearly.

3.2.1 Problem Identification and Motivation

The problem identification phase was undertaken using a combination of retrospective pathway analysis, operational observations, and literature synthesis. Local cancer waiting time reports and patient tracking data were reviewed to identify recurring delays between MDT outcomes, first oncology consultation, decision-to-treat, and treatment initiation. Findings were triangulated with evidence from the literature review, which demonstrated similar challenges across NHS organisations. This process established the need for an integrated operational analytics solution capable of identifying bottlenecks before breaches occur.

3.2.2 Objectives of a Solution

Solution objectives were derived through a requirements elicitation process combining literature-derived design requirements, operational needs identified through pathway review, and information requirements associated with national cancer waiting time standards. Requirements were grouped into data integration, operational monitoring, decision support, and usability domains.

3.3 Design and Development

The design and development stage focused on constructing an Integrated Data Repository (IDR) and a suite of operational decision-support dashboards. Development was undertaken iteratively using successive cycles of prototyping, evaluation and refinement with operational stakeholders.

The design and development activity involved two tightly coupled components:

1. Integrated Data Repository (IDR)
2. Decision-Support Dashboard

The IDR adopts a general-model architecture consisting of a staging layer for ETL and harmonisation, a central warehouse, and an application layer for analytics and visualisation, consistent with IDR literature for single-institution deployments (Gagalova *et al.*, 2020).

The dashboard is designed to support operational tasks rather than retrospective reporting, aligning with the role of operational dashboards in healthcare performance management (Buttigieg *et al.*, 2017).

Both components were developed iteratively, guided by explicit design principles derived from prior research and refined during prototyping.

3.3.1 Requirements Elicitation

Initial requirements were gathered through discussions with operational managers within Portsmouth Oncology Centre, review of existing reporting processes, and analysis of pathway-monitoring activities. Particular attention was paid to identifying operational questions that could not be answered efficiently using existing systems.

Key requirements included:

- identification of patients at risk of breaching national waiting-time standards;
- monitoring active oncology referrals and outstanding clinic activity;
- visibility of radiotherapy demand and capacity;
- identification of bottlenecks between referral and treatment initiation; and
- provision of near-real-time operational decision support.

3.3.2 Source System Analysis

Four primary operational datasets were identified, and three were used in the final artefact.

The oncology referral register consisted of a manually maintained SharePoint list used by administrative staff to track incoming referrals and clinical outcomes.

Oncology clinic activity was extracted from the Trust’s Patient Administrative System (PAS), providing appointment, booking and attendance data.

Radiotherapy operational data comprised referral, CT simulation and treatment events extracted from the local Oncology Information System - Aria for Radiation Oncology.

Due to additional projects involving the local Cancer Network, a reliable data source for SACT treatment is not available until January 2027, and was therefore excluded at this time, but design principles leave including SACT treatments at a later date.

The selected source systems reflected the operational stages of the post-MDT pathway. Together, these systems captured referral management, clinic scheduling, radiotherapy planning, and treatment delivery activities. However, each system was developed independently and used different data structures, identifiers, and reporting mechanisms. Consequently, substantial transformation and harmonisation activities were required before integrated pathway analysis could be performed.

3.3.3 Data Integration Strategy

A key design decision concerned the method by which operational data would be extracted from heterogeneous source systems. Alternative approaches including direct database connections, JSON-based exports, application programming interfaces (APIs), and manual reporting extracts were considered. However, functionality varied considerably between source systems. While some systems supported direct database access and others allowed JSON exports, comma-separated value (CSV) export functionality was available consistently across all source applications.

A CSV-based ingestion strategy was therefore adopted to maximise interoperability and minimise source-system dependencies. This approach allowed a common extraction process to be implemented across all datasets while avoiding the need for system-specific integration methods. The resulting architecture prioritised simplicity, portability, and maintainability over real-time integration, which was considered appropriate for an operational dashboard refreshed on a daily basis.

One exception was the Clinic appointment extract from PAS. Due to attempting to capture appointments booked, which acted as a proxy for triage, and also outcome, two reports were provided weekly in the form of Microsoft Excel workbooks containing separate Booking and Appointment worksheets. The revised ETL process automatically ingested both worksheets, preserving booking events and attendance outcomes independently before combining them within the intermediate integration layer. This approach improved representation of the underlying operational workflow while maintaining compatibility with the existing warehouse architecture.

3.3.4 ETL Architecture

Data extraction, transformation and loading processes were developed using Python and PostgreSQL. Individual ETL processes were created for each source dataset and orchestrated through an automated pipeline. The ETL layer was responsible for importing source extracts, performing data quality checks, standardising formats, and loading records into the staging layer of the IDR.

The architecture followed a layered design consisting of:

- Source systems;
- ETL and harmonisation processes;
- Staging tables;
- Intermediate integration views;
- Analytics-ready fact tables; and
- Power BI dashboards.

This separation of concerns simplified troubleshooting, improved transparency, and preserved data lineage throughout the repository.

The architecture deliberately separated raw operational data from analytical representations. Staging tables preserved source extracts in their original form, intermediate views applied business rules and data-quality controls, and fact tables provided analytics-ready datasets for dashboard consumption. This separation reduced coupling between source systems and reporting outputs while supporting auditability and future extension of the repository (See Figure 3.1).

3.3.5 Data Harmonisation and Business Rules

Data harmonisation presented a significant challenge due to variation in identifier quality, inconsistent recording practices, and differences in operational workflows between source systems.

NHS numbers were used as the primary identifier wherever possible and were standardised during transformation to remove formatting inconsistencies. Date formats were similarly harmonised to support accurate interval calculations across systems.

The oncology referral dataset presented particular challenges because it functioned as an operational activity log rather than a controlled referral register. Multiple records were frequently associated with the same oncology episode, reflecting administrative updates, onward referrals, and pathway exceptions. Intermediate transformation views were therefore developed to consolidate records and create a single analytical representation of each referral episode.

Additional business rules were implemented to identify external-provider referrals, duplicate records, deferred follow-up appointments, and non-standard pathways. These rules allowed

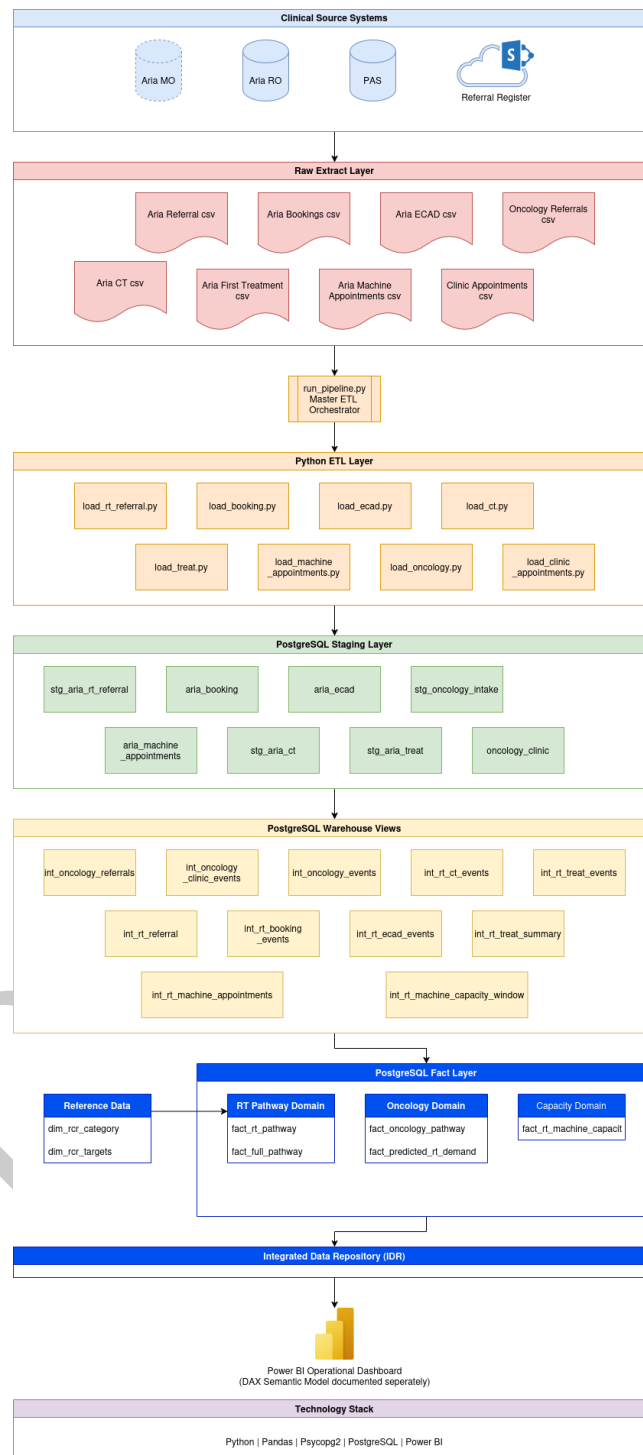


Figure 3.1: Logical Architecture of the Integrated Data Repository

operational reporting to focus on patients actively progressing through the pathway while preserving the underlying source data for audit purposes.

Following transformation and quality-control activities, analytics-ready fact tables were constructed for oncology pathways, radiotherapy pathways, and integrated referral-to-treatment pathways. These fact tables formed the data source for all dashboard visualisations and performance metrics.

Several previously unrecognised data-quality issues were identified during exploratory analysis. These included duplicate referral records, referrals represented by multiple administrative updates, delayed recording of clinic outcomes, and pathway records relating to patients managed by external providers. The identification and resolution of these issues was necessary to ensure that reported pathway metrics reflected true patient progression rather than administrative artefacts.

3.3.6 Software Engineering and Operational Design

Although the artefact was developed as a research prototype, several software-engineering practices were incorporated to improve reliability and maintainability.

Logging was implemented throughout the ETL process to support monitoring, troubleshooting, and auditability. All major extractions, transformation and loading activities produced structured log entries allowing operational issues to be investigated retrospectively.

Database credentials and configuration settings were externalised through environment variables rather than being embedded directly within the application code. This approach aligns with principles outlined within the Twelve-Factor App methodology, separating configuration from code and reducing the risk of credential exposure.

Together, these practices improved operational robustness and facilitated deployment across development and production environments without modification to application source code.

3.3.7 Pathway Data Model

The purpose of the pathway data model was to transform fragmented operational records into a series of standardised pathway events capable of supporting interval analysis. A key challenge identified during development was individual source systems recorded only portions of the overall oncology pathway. Consequently, no single source provided a complete end-to-end view from referral to treatment initiation.

The pathway model therefore adopted a canonical event-based approach. Individual operational events were mapped to a common set of pathway milestones including referral receipt, oncology triage, first clinic appointment, radiotherapy referral, CT simulation, and treatment commencement. These events were subsequently linked through patient identifiers and temporal matching logic to create an integrated representation of each patient pathway.

From these canonical events, a series of operational intervals were derived, including referral-to-triage, triage-to-clinic, clinic-to-radiotherapy referral, radiotherapy referral-to-CT simula-

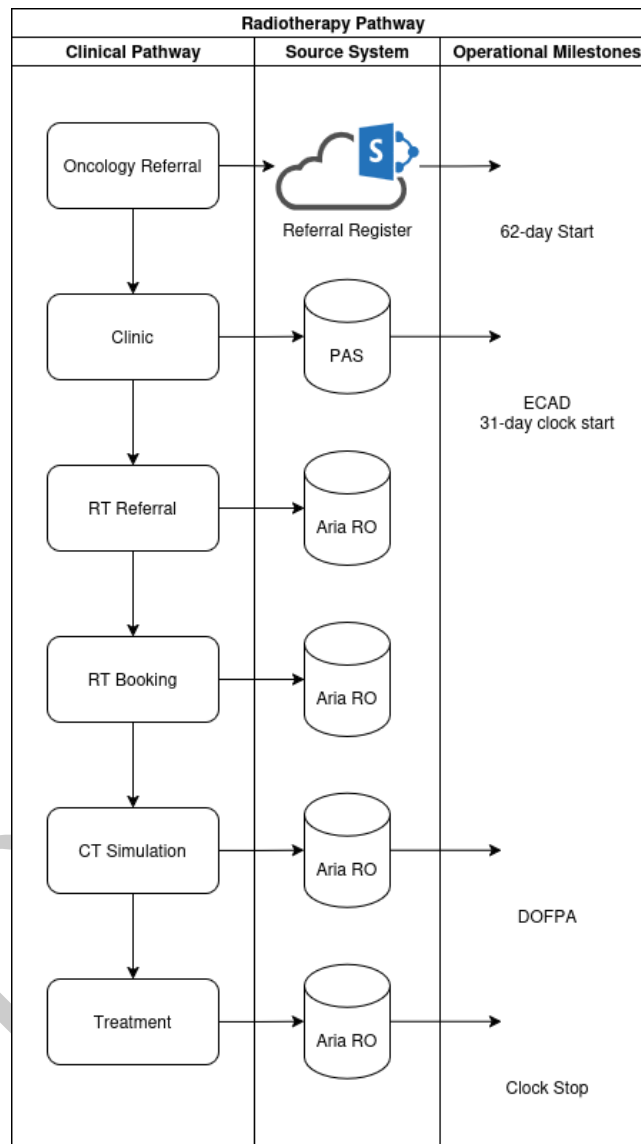


Figure 3.2: Radiotherapy Pathway, Source Systems and Performance Milestones

tion, and CT simulation-to-treatment. These intervals formed the basis of all dashboard metrics and pathway analyses.

The use of standardised pathway events aligned with Design Principle 3, ensuring that all metrics were derived consistently across source systems and remained traceable to their originating records. Figure 3.2 illustrates the relationship between pathway events, source systems, and derived performance milestones.

3.3.8 Dashboard Development

Three operational dashboards were developed using Microsoft Power BI. Each dashboard was designed to support a distinct operational management task while sharing a common data repository and data model.

The Oncology Referrals dashboard focused on referral management activities and provided visibility of active referrals, clinic waiting times, booking lead times, and unresolved clinic outcomes. The dashboard combined referral-register data with clinic activity to provide a current operational view of patients progressing through the oncology pathway.

The Radiotherapy Service Health dashboard focused on treatment delivery and capacity management. Operational metrics included active patients, expected future referrals, booking backlog, patient risk categorisation, and projected capacity utilisation. Particular emphasis was placed on presenting risk information alongside capacity and demand metrics to support operational intervention.

The Post-MDT Pathway Overview dashboard integrated data from both oncology and radiotherapy workflows to provide an end-to-end view of patient progression from referral to treatment. The dashboard visualised pathway intervals, operational bottlenecks, and trends in pathway performance over time.

A fourth dashboard addressing Systemic Anti-Cancer Therapy (SACT) pathways was identified during requirements elicitation. However, due to the lack of a reliable and stable SACT data source during the study period, this component was deferred for future development.

Rather than producing a single all-purpose dashboard, functionality was distributed across multiple focused dashboards. This approach aligned with ISO 9241-110 principles by supporting specific operational tasks while reducing visual complexity. Users were therefore presented with information relevant to individual operational decisions rather than a large collection of unrelated metrics.

3.3.9 Iterative Artefact Refinement

Artefact development followed an iterative process consistent with Design Science Research principles. Successive prototypes were demonstrated to operational stakeholders, with feedback being incorporated into later design iterations.

The first development cycle focused on radiotherapy operational management. Initial prototypes concentrated on patient tracking and waiting-time performance; however, stakeholder

feedback identified a need for greater visibility of future demand and available treatment capacity. This led to the development of capacity-demand forecasting visuals and booking risk indicators.

The second development cycle focused on oncology referral management. Early versions primarily reported referral volumes and clinic activity. User feedback indicated a stronger operational requirement for active waiting-list management, prompting the addition of patient-group classifications, current booked waits, and referral work-lists prioritised by waiting time.

The third development cycle focused on integrated pathway analysis. Stakeholders identified a need to understand where delays accumulated across the patient journey rather than within individual operational areas. This resulted in the development of pathway-stage metrics and a cumulative waterfall visual illustrating time spent between referral, triage, clinic attendance, radiotherapy referral, CT simulation, and treatment initiation.

These iterative refinements ensured that the artefact remained closely aligned with operational requirements while simultaneously providing evidence regarding the effectiveness of the proposed design principles.

3.3.10 Data Quality Management

Exploratory analysis identified several data-quality challenges including duplicate referral records, delayed administrative updates, missing appointment outcomes, and inconsistent coding practices. Rather than modifying source data directly, data-quality controls were implemented within the transformation layer of the repository.

Business rules were used to identify duplicate referral episodes, exclude non-standard pathways from operational waiting-time calculations, and classify clinician-directed surveillance appointments separately from active patient pathways. This approach preserved the fidelity of source data while improving the validity and interpretability of analytical outputs.

Data-quality findings were also treated as a research outcome in their own right, providing insight into the operational limitations of existing pathway-monitoring processes and informing recommendations for future pathway digitisation initiatives.

The discovery of these data-quality issues highlighted limitations in existing operational reporting processes. In particular, the distinction between administrative workflow events and true pathway events emerged as a recurring challenge. This finding informed the design of the pathway data model and represents an additional practical contribution of the study.

3.4 Design Principles

Design principles represent the design knowledge contribution of this study. They are formulated using a standard DSR structure: *context*, *intervention*, and *rationale*. These initial principles are refined following evaluation.

Artefact Component	Research Objective
IDR Architecture	Objective 3
Oncology Dashboard	Objective 4
RT Dashboard	Objective 4
Pathway Dashboard	Objective 1
Evaluation Activities	Objective 5
Design Principles	Objective 6

Table 3.1: Artefact to Research Objective mapping

DP1: Surface Breach Risks as a Leading Indicator

If operational teams must prioritise actions in time-constrained cancer pathways, then dashboards should surface leading indicators of breach risk (e.g. days remaining to breach thresholds), because prospective visibility enables earlier and more effective intervention than retrospective compliance metrics.

DP2: Pair Risk Information with Actionable Capacity Data

If users are expected to act on identified risks, then risk indicators should be presented alongside relevant capacity and demand information, because risk awareness without visible operational levers does not support decision-making.

DP3: Use Canonical, Standard-Aligned Pathway Events

If pathway metrics are to be trusted and comparable, then all intervals should be derived from canonical event definitions aligned to NCWTMDS rules, because inconsistent definitions undermine credibility and evaluation validity.

DP4: Match Data Refresh Cadence to the Action Horizon

If decisions depend on short operational windows, then data refresh frequency should be aligned to the decision horizon (e.g. daily or near-real-time where feasible), because stale information degrades decision quality.

DP5: Preserve Data Provenance and Auditability

If analytics are deployed in regulated healthcare environments, then all derived metrics must maintain transparent provenance to source systems, because trust, governance, and reconciliation are essential for adoption.

DP6: Align Interaction Design with Real Operational Tasks

If dashboards are intended to support complex operational work, then interaction design should reflect users' real task sequences, because task-aligned interfaces reduce cognitive load and improve decision quality (International Organization for Standardization, 2020).

These principles directly inform both the IDR architecture and the dashboard design, and serve as evaluation criteria in Section 3.5.

3.5 Evaluation Strategy

Evaluation combines formative and summative methods across four dimensions.

3.5.1 Usability Evaluation

- System Usability Scale (SUS)
- Heuristic review grounded in ISO 9241-110
- Think-aloud walkthroughs

3.5.2 Utility and Effectiveness

- Task-based experiment with operational users
- Primary measures:
 - time-to-answer key operational questions,
 - accuracy of breach-risk identification,
 - decision quality in constrained scheduling scenarios

3.5.3 Technical Validity

- Reconciliation of dashboard metrics with source SQL queries
- Validation of interval derivations against NCWTMDS rules

3.5.4 Design Knowledge Evaluation

- Synthesis of user feedback and performance outcomes
- Refinement and confirmation of final design principles

3.6 Ethical and Governance Considerations

The study operates within UK GDPR and NHS data governance requirements. All data are de-identified or pseudonymised, with access controls and separation between operational and research use. The project follows university ethics procedures and local Trust information governance guidance.

3.6.1 Research Limitations

- **Single-site study:** The artefact will be developed and evaluated within a single NHS Trust, which may limit generalisability.
- **Data quality and availability:** Findings depend on the completeness and accuracy of source systems.
- **Retrospective data:** Retrospective data may not fully represent future operational conditions.
- **Usability sample size:** Evaluation participants are likely to represent a relatively small operational user population.
- **Artificial evaluation environment:** Demonstration scenarios may not fully replicate live decision-making environments.
- **DSR limitation:** Evaluation assesses utility and usability rather than directly measuring improvements in patient outcomes.

3.7 Summary

This chapter has outlined a rigorous, design-science-based methodology that supports both artefact creation and theory-driven knowledge contribution. By embedding explicit design principles within the DSR framework and evaluating the artefact using multiple complementary methods, the study ensures methodological robustness, practical relevance, and academic contribution.

Chapter 4

Results and Analysis

4.1 Introduction

This chapter presents the results obtained from the development and application of the Integrated Data Repository (IDR) and decision-support dashboard suite described in Chapter 3. Following construction of the repository, integrated pathway data were analysed to identify operational bottlenecks, assess service demand and capacity, and evaluate the utility of the artefact for supporting operational decision-making.

Findings are presented according to the research objectives. The chapter begins by summarising repository development and data quality findings before presenting analysis of oncology referrals, radiotherapy operational performance, and end-to-end pathway delays. The chapter concludes by examining the extent to which the developed artefact supports the design principles established in Chapter 3.

4.2 Repository Development Results

The IDR successfully integrated datasets originating from multiple operational systems including oncology referral register, Patient Administration System (PAS), and the Aria radiotherapy management system. Data were extracted, transformed and loaded into a PostgreSQL-based repository using layered architecture consisting of staging, intermediate and analytical reporting layers.

Development of the repository identified several previously unrecognised data quality and interoperability challenges. These included duplicate oncology referral records, inconsistent representation of referral episodes, delayed administrative updates, and differences in event definitions between operational systems.

The use of intermediate integration views allowed business rules to be applied without modifying source data directly. This approach enabled the construction of analytics-ready fact tables capable of supporting operational reporting while preserving auditability and traceability to originating systems.

Oncology Referrals

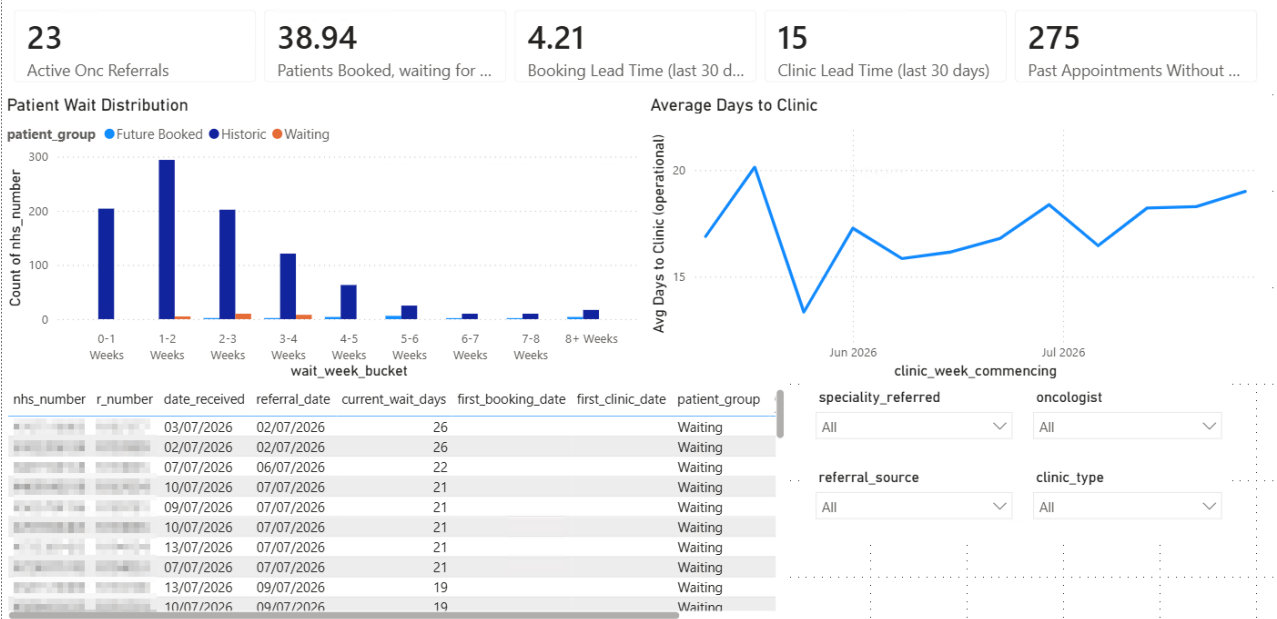


Figure 4.1: Operational Oncology Referral Dashboard

The repository subsequently became the single source of truth for all dashboard metrics and pathway analysis.

4.3 Oncology Referral Analysis

4.3.1 Referral Workload

Analysis of the oncology referral dashboard identified 23 active oncology referrals at the time of analysis. While this represents a manageable operational workload, variation was observed in waiting times between referral and clinic attendance.

The distribution of patient waits demonstrated that the majority of patients were clustered within the first three weeks following referral. Smaller numbers of patients remained on the waiting list beyond six weeks, suggesting that the majority of referrals progressed through the booking process relatively efficiently.

4.3.2 Booking and Clinic Lead Times

The average referral-to-booking interval during the previous 30 days was 4.15 days, indicating that referrals were generally triaged and allocated appointments within one working week.

The median referral-to-clinic interval was 15 days. Trend analysis demonstrated that this value remained relatively stable across the observation period, fluctuating between approximately 13 and 20 days.

These findings suggest that initial referral processing and appointment allocation were not the primary sources of pathway delay. The majority of patients received clinic appointments within a timeframe considerably shorter than national cancer waiting-time thresholds.

4.3.3 Future Booked Waits

Despite relatively short clinic lead times, the average wait experienced by patients already holding future appointments was 38.94 days.

This distinction highlights an important operational observation. The referral-to-clinic metric reflects patients who have already completed the clinic stage, while the booked waiting time reflects future demand against available clinic capacity. The higher booked wait therefore suggests increasing pressure on future appointment capacity rather than a deterioration in historic clinic performance.

4.3.4 Incomplete Outcome Recording

The dashboard identified 275 historical clinic appointments without recorded outcomes.

This finding represents a significant data-quality issue because pathway progression, attendance analysis, and downstream performance reporting depend upon accurate recording of appointment outcomes. The presence of a large number of unresolved appointments highlights the importance of integrated operational monitoring and demonstrates how the artefact can reveal process weaknesses that may not be visible through conventional reporting.

4.4 Radiotherapy Service Analysis

4.4.1 Operational Workload

The radiotherapy dashboard identified 204 active patients undergoing management within the radiotherapy pathway.

In addition, 32 future referrals were predicted to enter the service under a 62-day pathway, defined as oncology referrals booked into Clinical Oncology clinic. This provides operational managers with advance visibility of future demand and supports proactive planning of treatment capacity.

4.4.2 Capacity Utilisation

Capacity-demand modelling identified an average projected utilisation of 81.56% across the next 30 days, exceeding the departmental planning threshold of 80%. This threshold is used operationally to preserve flexibility for urgent treatments, machine downtime, replanning activity, and other unplanned service pressures.

Radiotherapy Service Health

Last load: 28/07/2026 11:19:21

204

Active Patients

32

Expected 62-day Referrals

81.56

Avg Utilisation (next 30 days)

138

Open Bookings

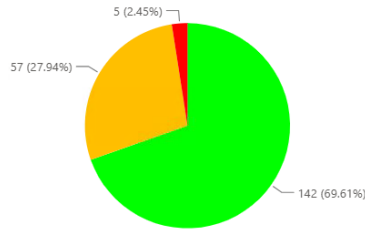
OncologySpeciality

All

oncologist

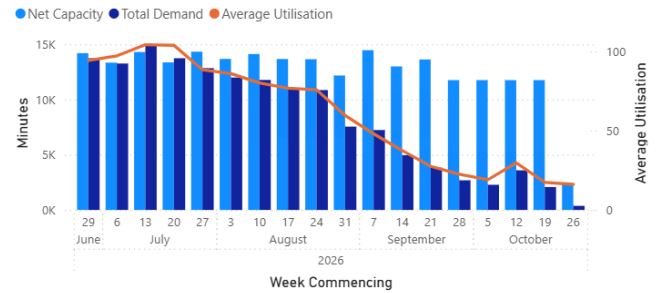
All

Active Patients by RAG Description



● Within JCCO Guidelines ● Outside JCCO Guideline (<31d) ● Over 31 Days from ECAD

Radiotherapy Capacity and Demand



NHS number	R number	RCR Category	Tumour	Oncologist	ECAD	Booked Start Date	ECAD to Treat	Activity note
123456789	123456789	CAT2	Prostate	Dr. Smith, J.	02/07/2026	26/08/26	55.00	CM SABR Prostate 5# Three times a week PT choice Scan before holiday treat after
987654321	987654321	CAT2	Breast	Dr. Jones, M.	02/07/2026	13/08/26	42.00	Pt choice, delay by week due to bereavement. AS R breast 26gy/5# must treat by 1
567890123	567890123	CAT2	Lower GI	Dr. Brown, K.	09/07/2026	17/08/26	39.00	JL Rectum 25# Concurrent Chemo Treat by 08/08/26
345678901	345678901	CAT2	Breast	Dr. White, L.	08/07/2026	10/08/26	33.00	DO NOT MOVE- PATIENT TO BE HELPED ONTO A BED AND THEN PATSLIDE ONTO surgery)
234567890	234567890	CAT2	Non-melanoma skin cancer	Dr. Green, P.	26/06/2026	28/07/26	32.00	SR - Left Ear Pinna - 50Gy 20#
123456789	123456789		Breast	Dr. White, L.	07/07/2026	04/08/26	28.00	RA R/BREAST 5# ECAD 7/7 TRT BY 6/8
987654321	987654321		Breast	Dr. White, L.	07/07/2026	04/08/26	28.00	FAST FORWARD BOOST - FOR DAILY BREAST CBCT IMAGING - PLEASE ENSURE CC
Total								

Figure 4.2: Radiotherapy Capacity and Demand Dashboard

Forecast utilisation declined across the later periods of the planning horizon. However, this reduction should be interpreted with caution because future demand is inherently less complete than near-term demand. The model includes patients currently booked for treatment and referrals that have already entered the radiotherapy pathway, but does not yet reflect patients who have not progressed sufficiently through upstream stages of the pathway. Consequently, the later forecasting periods are likely to underestimate true future demand.

Despite this limitation, the model provides a useful indication of current service pressure and demonstrates that the radiotherapy service is operating at or slightly above its internal planning threshold.

4.4.3 Patient Risk Stratification

Patient-level risk analysis identified:

- 142 patients (69.61%) within JCCO guidelines (Ash *et al.*, 2004).
- 57 patients (27.94%) outside JCCO guidelines, but within NCWT targets (Johnson, 2024).
- 5 patients (2.45%) outside the NCWT 31-day standard.

Although the majority of patients remained within the recommended timeframes, the presence of patients approaching or exceeding target thresholds demonstrates the value of a proactive monitoring approach.

Rather than identifying breaches retrospectively, the dashboard allows operational teams to focus attention on patients most likely to experience future delays.

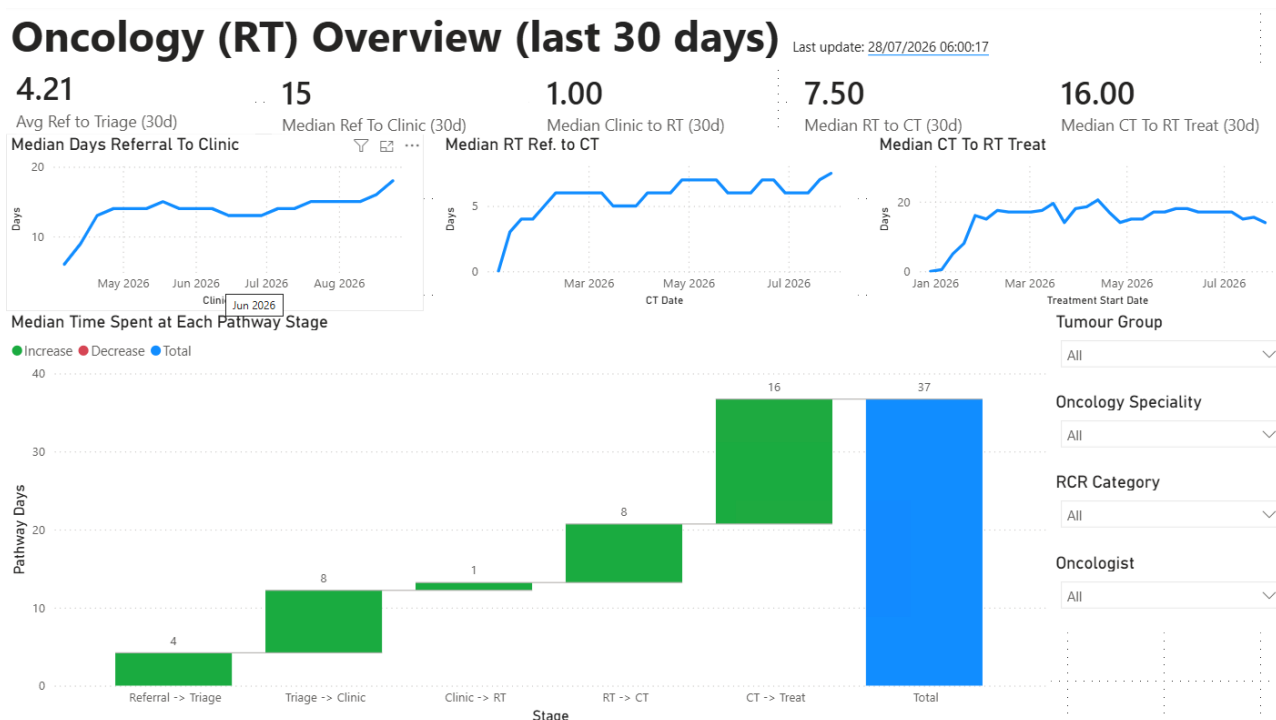


Figure 4.3: Integrated Post-MDT Pathway Dashboard

4.5 End-to-End Pathway Analysis

4.5.1 Pathway Duration

The integrated pathway dashboard enabled analysis of patient progression from referral through to treatment initiation.

The median pathway duration was 36 days and was composed of five principal operational stages:

Stage	Days
Referral →Triage	4
Triage →Clinic	8
Clinic →RT Referral	1
RT Referral →CT	8
CT →Treatment	16
Total	37

4.5.2 Bottleneck Identification

The largest delay occurred between CT simulation and treatment commencement, accounting for 16 days of the overall pathway.

The second largest delay occurred between triage and first clinic attendance, accounting for 8 days.

Together these two stages represented approximately two-thirds of the total pathway duration, suggesting that operational improvement efforts should be concentrated in these areas.

In contrast, the interval between clinic attendance and radiotherapy referral was only one day, indicating that MDT and consultant decision-making processes were not a significant source of delay.

4.5.3 Temporal Trends

Trend analysis demonstrated that:

- Referral-to-clinic performance remained relatively stable.
- RT Referral-to-CT increased from approximately 0 days to 7 days before stabilising.
- CT-to-Treatment remained consistently between 15 and 20 days.

The persistence of the CT-to-treatment interval suggests that a substantial proportion of pathway time is spent between CT simulation and treatment initiation. However, this interval represents an aggregation of several planning and preparation activities rather than a single operational process.

Following CT simulation, patients typically undergo contouring, treatment planning, plan review, quality assurance, and treatment scheduling before treatment can commence. These activities do not currently form part of the canonical pathway events recorded within the repository but nevertheless consume operational time and resources.

Consequently, further investigation is required before this interval can be interpreted solely as a treatment-capacity issue. Opportunities for future work are discussed in Chapter ??.

4.5.4 Research Question 1

The pathway dashboard directly addressed Research Question 1:

Where do delays occur between MDT outcome and treatment?

Analysis demonstrated that delays are concentrated primarily within:

1. CT Simulation → Treatment Start.
2. Triage → Clinic Attendance.

These stages therefore represent the most significant opportunities for operational intervention.

Design Principle	Evidence from Artefact
DP1: Surface Breach Risks	RAG categorisation and patient risk work-lists
DP2: Pair Risk with Capacity	Capacity and demand dashboard visualisations
DP3: Canonical Events	Standardised pathway intervals derived from warehouse fact tables
DP4: Refresh Cadence	Automated ETL pipeline and scheduled refresh processes
DP5: Provenance and Auditability	Staging, intermediate and fact-layer architecture with ETL logging
DP6: Task-Oriented Design	Separate referral, radiotherapy and pathway dashboards aligned to operational workflows

4.6 Evaluation of Design Principles

The evaluation suggests that all six design principles were successfully implemented within the final artefact.

Particularly strong support was observed for DP1 and DP2. Operational users were able to identify patients approaching target breaches while simultaneously assessing available capacity to address identified risks. This combination moved reporting from retrospective performance measurement towards proactive operational management.

4.7 Summary

The developed IDR successfully integrated multiple operational datasets and enabled detailed analysis of post-MDT oncology workflows. Repository development exposed several data quality challenges while establishing a unified analytical platform for operational decision support.

Analysis of oncology referrals demonstrated relatively efficient referral processing but highlighted growing pressure on future clinic capacity. Radiotherapy analysis identified manageable overall utilisation but a small cohort of higher-risk patients requiring ongoing monitoring.

End-to-end pathway analysis demonstrated that the greatest delays occur between CT simulation and treatment initiation, with an additional contribution arising between triage and first clinic attendance. These findings provide evidence-based targets for future service improvement initiatives and demonstrate the practical utility of the developed artefact for operational oncology management.

References

- Ash, D., Barrett, A., Hinks, A. and Squire, C. (2004) ‘Re-audit of radiotherapy waiting times 2003,’ *Clinical Oncology*, 16(6), pp. 387–394, URL <http://dx.doi.org/10.1016/j.clon.2004.06.006>
- Butterworth, Victoria, Young, Tom, Drake, Haleema, Palmer, Isabel, Avgouleas, Tania, Ivy, Eleanor, Andriolo, Joshua, Creppy, Carole, Routledge, Corla, Adjogatse, Delali, Kong, Anthony, Petkar, Imran, Reis Ferreira, Miguel, Eaton, David, Lei, Mary, Mission, Sarah, Vilic, Dijana and Guerrero Urbano, Teresa (2025) ‘Data-centric Artificial Intelligence and Cancer Research: Construction of a Real-World Head and Neck Treatment Data Repository,’ *ESMO Real World Data and Digital Oncology*, 9(C), URL <http://dx.doi.org/10.1101/2025.02.12.25322092>
- Buttigieg, Sandra C., Pace, Adriana and Rathert, Cheryl (2017) ‘Hospital performance dashboards: A literature review,’ *Journal of Health Organization and Management*, 31(3), pp. 385–406, URL <http://dx.doi.org/10.1108/JHOM-04-2017-0088>
- Calman, Kenneth and Hine, Deirdre (1995) *A Policy Framework for Commissioning Cancer Services : A Report by the Expert Advisory Group on Cancer to the Chief Medical Officers of England and Wales*, Tech. rep., Department of Health, London
- Chakravarty, PD, Ton, T, Scott, A, Doherty, C, Douglas, CM and Montgomery, J (2023) ‘Outpatient secondary care pathways for head and neck cancer referral result in patient delays for cancer treatment,’ *The Annals of The Royal College of Surgeons of England*, 105(4), pp. 352–356, URL <http://dx.doi.org/10.1308/rcsann.2022.0111>
- Chang, Jee Suk, Kim, Hyunwook, Baek, Eun Sil, Choi, Jeong Eun, Lim, Joon Seok, Kim, Jin Sung and Shin, Sang Joon (2025) ‘Continuous multimodal data supply chain and expandable clinical decision support for oncology,’ *npj Digital Medicine*, 8(1), p. 128, URL <http://dx.doi.org/10.1038/s41746-025-01508-2>
- Coca-Pelaz, Andrés, Takes, Robert P., Hutcheson, Kate, Saba, Nabil F., Haigentz, Misak, Bradford, Carol R., de Bree, Remco, Strojan, Primož, Lund, Valerie J., Mendenhall, William M., Nixon, Iain J., Quer, Miquel, Rinaldo, Alessandra and Ferlito, Alfio (2018) ‘Head and Neck Cancer: A Review of the Impact of Treatment Delay on Outcome,’ *Advances in Therapy*, 35(2), pp. 153–160, URL <http://dx.doi.org/10.1007/s12325-018-0663-7>

- Conway, Alicia-Marie, Concannon, Matthew, Brown, Steven, Hussain, Humera, Marland, April, Griffiths, Matthew, Plummer, Barney, Warrington, Kelly, Bridges, Claire, Lightning-Jones, Karen and Cook, Natalie (2025) ‘Overcoming data management challenges in oncology research: Lessons from an NHS, industry, technology start-up and academic collaboration,’ *European Journal of Cancer*, 228, p. 115705, URL <http://dx.doi.org/10.1016/j.ejca.2025.115705>
- Dawson, Christian W. (2015) *Projects in Computing and Information Systems: A Student's Guide*, 3rd ed., Pearson, Harlow
- Gagalova, Kristina K., Elizalde, M. Angelica Leon, Portales-Casamar, Elodie and Görges, Matthias (2020) ‘What You Need to Know Before Implementing a Clinical Research Data Warehouse: Comparative Review of Integrated Data Repositories in Health Care Institutions,’ *JMIR Formative Research*, 4(8), p. e17687, URL <http://dx.doi.org/10.2196/17687>
- Guilbert, Baptiste, Rieu, Tristan, Delamarre, Denis, Laurent, Frédéric, Serre, Fanny, Leroy, Sarah, Paillard, Elliott, Le Buhan, Nolwenn, Ribault, Alain and Cuggia, Marc (2024) ‘ONCOFAIR Project: Improving Data Interoperability in Oncology Chemotherapy Treatments for Data Reuse,’ in Mantas, John, Hasman, Arie, Demir, George, Saranto, Kaija, Marschollek, Michael, Arvanitis, Theodoros N., Ognjanović, Ivana, Benis, Arriel, Gallos, Paris, Zoulias, Emmanouil and Andrikopoulou, Elisavet (eds.), ‘Studies in Health Technology and Informatics,’ *Studies in Health Technology and Informatics*, vol. 316, IOS Press, pp. 1373–1377, URL <http://dx.doi.org/10.3233/SHTI240667>
- Han, Xudong, Gao, Xianglun, Qu, Xiaoyi and Yu, Zhenyu (2025) ‘Multi-Agent Medical Decision Consensus Matrix System: An Intelligent Collaborative Framework for Oncology MDT Consultations,’ URL <http://dx.doi.org/10.48550/arXiv.2512.14321>
- Hanna, Timothy P., King, Will D., Thibodeau, Stephane, Jalink, Matthew, Paulin, Gregory A., Harvey-Jones, Elizabeth, O’Sullivan, Dylan E., Booth, Christopher M., Sullivan, Richard and Aggarwal, Ajay (2020) ‘Mortality due to cancer treatment delay: Systematic review and meta-analysis,’ *BMJ*, 371, p. m4087, URL <http://dx.doi.org/10.1136/bmj.m4087>
- Haward, Robert A. (2006) ‘The Calman–Hine report: A personal retrospective on the UK’s first comprehensive policy on cancer services,’ *The Lancet Oncology*, 7(4), pp. 336–346, URL [http://dx.doi.org/10.1016/S1470-2045\(06\)70659-3](http://dx.doi.org/10.1016/S1470-2045(06)70659-3)
- HealthOrbit (2025) ‘AI Solutions for Real Clinical Workflows | HealthOrbit AI,’ <https://healthorbit.ai/>
- Hevner, Alan R, March, Salvatore T, Park, Jinsoo and Ram, Sudha (2004) ‘Design Science in Information Systems Research,’ *MIS Quarterly*, 28(1)

- Hunkirchen, Peter () ‘Guidelines and Standards,’ <https://www.usability-ux.fit.fraunhofer.de/en/guidelines-and-standards.html>
- International Organization for Standardization (2020) ‘Ergonomics of human-system interaction Part 110: Interaction principles,’
- Johnson, Peter (2024) ‘Updating the NHS cancer waiting time standards,’ *BMJ Oncology*, 3(1), URL <http://dx.doi.org/10.1136/bmjonc-2024-000337>
- Jung, Hyun Ae, Jeong, Oksoon, Chang, Dong Kyung, Park, Sehhoon, Sun, Jong-Mu Sun, Lee, Se-Hoon, Ahn, Jin Seok, Ahn, Myung-Ju and Park, Keunchil (2021) ‘Real-time automatically updated data warehOuse in healThcare (ROOT): An innovative and automated data collection system - Jung- Translational Lung Cancer Research,’ *Translational Lung Cancer Research*, 10(10), pp. 3865–3874, URL <http://dx.doi.org/10.21037/tlcr-21-531>
- Law, Nicholas Lee Wei, Hong, Lai Wei, Tan, Shirley Siang Ning, Foo, Chuan Jie, Lee, Daniel and Voon, Pei Jye (2024) ‘Barriers And Challenges Of Multidisciplinary Teams In Oncology Management: A Scoping Review Protocol,’ *BMJ Open*, 14(2), p. e079559, URL <http://dx.doi.org/10.1136/bmjopen-2023-079559>
- Lim, Alison E., Williamson, Andrew, Green, Freddie, Lee, Ying Ki, Li, Lucy, Moen, Christy, Vasanthan, Rishi, Wharf, Olivia, Wong, Jeremy, Paleri, Vinidh and Network), INTEGRATE (The UK ENT Trainee Research (2025) ‘Clinical Factors Influencing the Compliance With National Head and Neck Cancer Targets in the United Kingdom: Results From a National Cohort Study,’ *Clinical Otolaryngology*, 50(4), pp. 739–750, URL <http://dx.doi.org/10.1111/coa.14314>
- Liu, Qicai, Hu, Zhichao, Huang, Tao, Niu, Yupeng, Zhang, Xinche, Ma, Shanwu, Lin, Chutong, Huat, Goh Kim, Kwon, Hyeokkoo Eric, Gao, Feng, Sun, Xianfu, Ying, Zhitao and Qiang, Guangliang (2026a) ‘EvoMDT: A self-evolving multi-agent system for structured clinical decision-making in multi-cancer,’ *npj Digital Medicine*, 9(1), p. 124, URL <http://dx.doi.org/10.1038/s41746-025-02304-8>
- Liu, Yunyou, Wang, Fei, Wang, Peng, Zhou, Zhen, Wang, Hongqian, Li, Jingyao, Qiu, Yang, Wang, Haidong and Miao, Siwei (2026b) ‘AI-MDT: An automatic and intelligent multidisciplinary team consultations platform for lung cancer diagnosis,’ *Journal of Cancer Research and Clinical Oncology*, 152(1), p. 32, URL <http://dx.doi.org/10.1007/s00432-025-06413-5>
- Morris, E, Haward, R A, Gilthorpe, M S, Craigs, C and Forman, D (2006) ‘The impact of the Calman–Hine report on the processes and outcomes of care for Yorkshire’s colorectal cancer patients,’ *British Journal of Cancer*, 95(8), pp. 979–985, URL <http://dx.doi.org/10.1038/sj.bjc.6603372>

- Morris, E., Haward, R. A., Gilthorpe, M. S., Craigs, C. and Forman, D. (2008) 'The impact of the Calman-Hine report on the processes and outcomes of care for Yorkshire's breast cancer patients,' *Annals of Oncology*, 19(2), pp. 284–291, URL <http://dx.doi.org/10.1093/annonc/mdm432>
- NHS Digital (2026) 'Cancer Waiting Times Data Collection (CWT),' <https://digital.nhs.uk/data-and-information/data-collections-and-data-sets/data-collections/cancerwaitingtimescwt>
- NHS Wales (2024) *Multidisciplinary Team (MDT) Charter*, Tech. rep., NHS Wales Cancer Network
- Parker, Jade (2026) 'Six ways to improve cancer MDT meetings,'
- Peffer, Ken, Tuunanen, Tuure, Rothenberger, Marcus A. and Chatterjee, Samir (2007) 'A design science research methodology for information systems research,' *Journal of Management Information Systems*, 24(3), pp. 45–77, URL <http://dx.doi.org/10.2753/MIS0742-1222240302>
- Price, G, Roche, M and Wight, R (2014) 'The impact of head and neck cancer service guidance on surgical workload,' *The Bulletin of the Royal College of Surgeons of England*, 96(9), pp. 300–303, URL <http://dx.doi.org/10.1308/147363514X14042954769195>
- RCR (2024) *Radiotherapy Briefing - May 2024*, Briefing Paper, Royal College of Radiologists
- Saleh, Mona, Kaplowitz, Elianna, Gellman, Caroline, Flint, Matthew, Holtzman, Sharonne, Mercier, Ann Marie and Kolev, Valentin (2024) 'Diagnosis-to-treatment interval in cervical cancer: An exploratory analysis of the SEER database,' *Gynecologic Oncology*, 190, pp. S361–S362, URL <http://dx.doi.org/10.1016/j.ygyno.2024.07.541>
- Sharma, Sonam, Bekelman, Justin, Lin, Alexander, Lukens, J. Nicholas, Roman, Benjamin R., Mitra, Nandita and Swisher-McClure, Samuel (2016) 'Clinical impact of prolonged diagnosis to treatment interval (DTI) among patients with oropharyngeal squamous cell carcinoma,' *Oral Oncology*, 56, pp. 17–24, URL <http://dx.doi.org/10.1016/j.oraloncology.2016.02.010>
- Soltan, A. A. S., Patel, S. and Sharma, J. (2025) '296P The TrustedMDT multi-agent ai copilot for tumour boards: Protocol for a two-phase evaluation of performance, safety, and feasibility,' *ESMO Real World Data and Digital Oncology*, 10, p. 100492, URL <http://dx.doi.org/10.1016/j.esmorw.2025.100492>
- Soukup, Tayana, Lamb, Benjamin W, Morbi, Abigail, Shah, Nisha J, Bali, Anish, Asher, Viren, Gandamihardja, Tasha, Giordano, Pasquale, Darzi, Ara, Sevdalis, Nick and

Green, James S A (2022) ‘Cancer multidisciplinary team meetings: Impact of logistical challenges on communication and decision-making,’ *BJS Open*, 6(4), p. zrac093, URL <http://dx.doi.org/10.1093/bjsopen/zrac093>

The Royal Wolverhampton NHS Trust (2024) *Cancer Operational Policy*, Policy OP03, The Royal Wolverhampton NHS Trust

Varma, Raghav, Mayne, Emily, Parry, Marina A., Nathan, Arjun, van der Meulen, Jan, Bahl, Amit, Stewart, Grant D. and Cowling, Thomas E. (2026) ‘Comparing the national cancer waiting times dataset with other linked electronic health records to measure treatment timeliness: A national cohort study of kidney cancer in England,’ *Cancer Epidemiology*, 102, p. 103030, URL <http://dx.doi.org/10.1016/j.canep.2026.103030>

Villemure-Poliquin, Noémie, Fu, Rui, Gaebe, Karolina, Kwon, Jin, Cohen, Marc, Ruel, Marianne, Ayoo, Kennedy, Bailey, Andrew, Galapin, Madette, Hallet, Julie and Eskander, Antoine (2025) ‘Delayed diagnosis to treatment interval (DTI) in head & neck cancers – A systematic review and meta-analysis,’ *Oral Oncology*, 160, p. 107106, URL <http://dx.doi.org/10.1016/j.oraloncology.2024.107106>

List of Acronyms

AI Artificial Intelligence

CDSS Clinical Decision Support System

CDW Clinical Data Warehouse

CWT Cancer Wait Time

DSR Design Science Research

DSRM Design Science Research Methodology

DTI Diagnosis to Treatment Interval

DTT Decision to Treat to Treatment

ETL Extract-Transform-Load

EPR Electronic Patient Record

FDS Faster Diagnosis Standard

FHIR Fast Healthcare Interoperability Resources

HCI Human Computer Interaction

HNC Head and Neck Cancer

HNSCC Head and Neck Squamous Cell Carcinoma

HR Hazard Ratio

IDR Integrated Data Repository

MDT Multidisciplinary Team

NCWTMDS National Cancer Waiting Times Monitoring Data Set

NHS National Health Service

NLP Natural Language Processing

PACS Picture Archiving and Communication System

PAS Patient Administration System

PTL Patient Tracking List

RCR Royal College of Radiologists

RIS (Radiology Information System)

RTT Referral To Treatment

DRAFT